

TEST PLAN

Digital Wallet Transfer Limits & Account Tiers

Version: 1.0

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1. Objective

The objective of this test plan is to validate that Digital Wallet transfer limits are enforced correctly across all account tiers, transfer types, and transaction states while ensuring financial integrity, regulatory compliance, security, and a consistent customer experience.

The strategy prioritizes identifying high-risk defects—particularly those that could allow users to exceed configured limits, incorrectly reduce available limits, or create inconsistent financial records—before validating lower-risk functionality.

2. Scope

In scope

- Tier-based transfer limits
- Daily transfer limits
- Daily reset logic
- Wallet-to-wallet transfers
- Wallet-to-bank transfers
- Own linked bank account exemption
- Pending transfers
- Failed transfers
- Reversed transfers
- Scheduled transfers
- Tier upgrades
- Error messaging
- Concurrent transfers

Out of scope

- Deposit limits
- International transfers
- Currency conversion
- Merchant payments
- Bank settlement processing
- Chargeback handling
- KYC verification workflow (except its impact on tier changes).

3. Test Objectives

Validate that:

- Transfer limits are correctly enforced.
- Daily limits are accurately calculated.
- Limits reset correctly.
- Pending, failed, and reversed transfers update limits correctly.
- Own linked bank accounts correctly bypass limits.
- Tier upgrades are immediately reflected.
- Scheduled transfers behave correctly.
- Concurrent transfers cannot bypass limits.
- Users receive meaningful error messages.
- Financial integrity is maintained under normal and edge-case conditions.

4. Test Assumptions

ID	Assumption
A1	All transfer amounts are denominated in Nigerian Naira (₦).
A2	Daily limits apply per sender wallet account, not per device or session.
A3	Scheduled transfers execute automatically without requiring user interaction.
A4	Transfer limit validation occurs before funds are debited.
A5	Transfer statuses include Pending, Successful, Failed, and Reversed.
A6	Every transfer has a unique transaction ID for idempotency.
A7	Users may initiate transfers from multiple devices simultaneously.

5. Test Environment

- QA Environment
- Staging APIs
- Scheduler enabled
- Database access for validation

- Feature flags enabled
- Mock bank gateway
- Ability to manipulate system/server time
- API logs enabled
- Audit logging enabled

6. Test Data Strategy

The following data sets will be prepared:

User Accounts

- Tier 1 users
- Tier 2 users
- Tier 3 users

Bank Accounts

- Linked account
- Unlinked account
- Invalid account

Wallet States

- No transfers
- Partial daily usage
- Daily limit exhausted
- Pending transfers
- Failed transfers
- Reversed transfers

Tier Scenarios

- User upgrading during transfer
- User upgrading before scheduled execution

Device Scenarios

- Single device
- Two devices
- Multiple simultaneous sessions

7. Open Questions for the Product Team

The following ambiguities, contradictions, and unhandled edge cases were identified during specification analysis. These should be resolved before detailed test case design begins, as several scenarios materially change what “correct” behaviour means.

#	Area	Ambiguity / Open Question	Example / Scenario	Risk / Why It Matters
1	Daily Limit Reset	Which timezone determines midnight — user local, server, or country?	User travels from Nigeria to the UK. When does the limit reset?	Users may receive inconsistent daily limits depending on location.
2	Pending Transfers Across Midnight	Does a transfer still Pending at midnight count against yesterday's limit, today's, or both?	₦40,000 sent at 11:59 PM remains pending until 12:05 AM.	Incorrect limit calculations; unfair rejection/approval of future transfers.
3	Reversal After Midnight	If a transfer is reversed after the daily reset, which day's limit is restored?	Transfer sent before midnight is reversed at 12:15 AM.	Daily limit may be restored incorrectly.
4	Atomicity / Midnight Cutover	Is the reset instantaneous, or is there a grace period? Which timestamp governs — request time, validation time, or processing time?	Request submitted 23:59:59.900, processed 00:00:00.200 due to load.	Inconsistent approvals/rejections and customer disputes; tests timing, concurrency, distributed behaviour.
5	Concurrent Transfers (Race Condition)	How should simultaneous transfers sharing the same remaining limit be handled?	Two ₦20,000 transfers submitted at once when only ₦20,000 remains.	Both could be approved, exceeding the daily limit — high fraud risk.
6	Scheduled Transfer Validation	Are limits validated at scheduling time or at 8:00 AM execution time?	User schedules ₦100,000 today, then spends the remaining limit before execution.	Unexpected failures or incorrect approvals.
7	Scheduled Transfers vs Daily Limit	If the daily limit is exhausted before execution, should the transfer fail, retry, or remain pending?	Daily limit is ₦0 before the scheduled execution.	Inconsistent UX and unclear business behaviour.
8	Tier Upgrade for Scheduled Transfers	Does a scheduled transfer use the tier at scheduling time or execution time?	Tier 1 user schedules ₦100,000, upgrades to Tier 2 before execution.	Different outcomes depending on implementation.
9	Tier Upgrade During Pending Transfer	If KYC completes while a transfer is pending, does the new tier apply immediately?	User upgrades while a transfer awaits processing.	Inconsistent enforcement of transfer limits.
10	Immediate Tier Upgrade	What does “takes effect immediately” mean — only new transfers, or pending/scheduled too?	User upgrades while multiple transactions are in progress.	Different system components may apply different rules.
11	Tier Downgrade	Can users be downgraded? What happens to scheduled or pending transfers if so?	User loses KYC status after scheduling a transfer.	Business rules are undefined.
12	Own Linked Bank Account Verification	How is “own linked account” verified — BVN, ownership check, or manual linking?	User links multiple bank accounts.	Incorrect users may bypass transfer limits.
13	Linked Account Status Change	What happens if a linked account is removed before a scheduled transfer executes?	Account unlinked after scheduling but before execution.	Transfer may incorrectly remain exempt from limits.

#	Area	Ambiguity / Open Question	Example / Scenario	Risk / Why It Matters
14	Linked Account Exemption Scope	Does exemption apply to daily and per-transaction limits, and to pending/scheduled transfers?	Large transfer to a linked account.	Rule interpretation may differ across services.
15	Failed Transfers Definition	What qualifies as “failed” — timeout, network failure, bank rejection, cancellation?	API timeout before confirmation.	Daily limit may not be restored correctly.
16	Partial Processing	What happens if the wallet debit succeeds but the bank credit fails?	Wallet debited, bank credit unsuccessful.	Incorrect balance and daily limit calculations.
17	Error Message Format	How is the remaining limit displayed — currency format, localization, decimals?	“Remaining Limit: ₦10,000” vs “10000”.	Poor UX and inconsistent messaging.
18	Scheduled Execution Tolerance	Does execution occur exactly at 8:00 AM, or is a delay window acceptable?	Scheduler starts at 8:03 AM due to load.	Users may question delayed transactions.
19	Multiple Scheduled Transfers	If several scheduled transfers exceed the remaining limit, which should succeed?	Three scheduled transfers compete for the same limit.	Non-deterministic execution order.
20	Concurrent Scheduled Jobs	How does the scheduler prevent multiple worker nodes from exceeding limits?	Multiple worker nodes process jobs concurrently.	Race conditions and limit breaches.
21	Tier 3 Maximum Transfer	Tier 3 has no per-transaction limit — is there a hidden system maximum?	User attempts to transfer ₦50,000,000.	Database or API limits may be exceeded unexpectedly.
22	Decimal Values	Are transfers allowed in kobo (decimals), or whole naira only?	₦10,000.50 transfer.	Validation inconsistencies across channels.
23	Daily Limit Boundary	Is a transfer equal to the exact remaining daily limit allowed?	Remaining limit ₦20,000; transfer amount ₦20,000.	Boundary rule must be explicitly defined.
24	Multiple Devices / Sessions	What happens if the same user initiates transfers from two devices simultaneously?	Mobile and tablet submit transfers at the same time.	Potential limit bypass through concurrent sessions.
25	Retry Mechanism	If a scheduled transfer fails on a temporary issue, does the system retry automatically?	Temporary network outage at 8:00 AM.	Unclear user expectations and inconsistent behaviour.

8. Risk Assessment and Prioritization

8.1 Risk priority levels

Priority	Meaning	Business Impact
PO – Critical	Core functionality; failure risks financial loss, security or compliance violations, or blocks usage.	Release blocker

Priority	Meaning	Business Impact
P1 – High	High-impact functionality affecting many users or key business processes; workarounds may exist.	Should be fixed before release
P2 – Medium	Important features with moderate impact; affects usability or specific scenarios, not core financial integrity.	Can follow in a subsequent release
P3 – Low	Cosmetic or minor usability issues, low-impact edge cases.	Low urgency
P4 – Very Low	Nice-to-have improvements with minimal customer impact.	Future backlog

8.2 Feature risk assessment

Feature	Priority	Why
Transfer Limit Validation (per transaction)	P0	Prevents customers exceeding regulatory limits; failure risks financial and compliance violations.
Daily Transfer Limit Calculation	P0	Incorrect calculation may allow excessive transfers or wrongly block legitimate users.
Concurrent Transfers	P0	Non-atomic handling of parallel requests can bypass limits entirely — high fraud risk.
Daily Limit Reset	P0	Incorrect reset logic affects every customer's available balance for the day.
Refund / Reversal Limit Adjustment	P0	Incorrect handling can permanently reduce, or incorrectly restore, a customer's limit.
Own Linked Bank Account Exemption	P1	Incorrect ownership validation may allow fraudulent bypass of limits.
Tier Upgrade After KYC	P1	Customers expect immediate effect; delays or inconsistency reduce trust.
Scheduled Transfers	P2	Important, but affects fewer users than real-time transfers.
Error Messages	P2	Users should understand rejections, but messaging is secondary to correct enforcement.
UI Formatting	P3	Cosmetic only.
Minor Visual Improvements	P4	No material business impact.

9. What to Test First, and Why

Testing priority follows financial and trust risk, not the order features appear in the specification. The goal is to surface anything that lets money silently exceed a stated limit, or silently vanish, before anything else is tested.

9.1 P0 — Critical (tested first)

Transfer limit validation

- Tier 1, Tier 2, and Tier 3 limits, including boundary values (limit – 1, exact limit, limit + 1).
- Wallet-to-wallet and wallet-to-bank transfers.

Why first: *this is the core business rule. If users can exceed limits, or are wrongly blocked, the feature has failed at its primary purpose.*

Daily transfer limit calculation

- Multiple transfers accumulating toward the daily limit.
- Exact daily limit reached; transfers attempted after exhaustion.

Why first: *a single calculation error propagates to every transfer and every customer.*

Concurrent transfers

- Multiple transfers submitted simultaneously against the same remaining limit.
- Two devices / two sessions racing the same account.

Why first: *race conditions are the highest fraud-risk gap — a non-atomic check-then-debit lets a user bypass limits simply by racing requests.*

Daily reset

- Transfers immediately before and immediately after midnight.
- Pending transfers that straddle the reset; timezone handling.

Why first: *incorrect reset logic silently corrupts every customer's daily allowance, not just one transaction.*

Refund and reversal processing

- Successful reversal, partial reversal, failed transfer, duplicate reversal.

Why first: *incorrect accounting either unfairly blocks customers or grants them extra headroom they shouldn't have.*

9.2 P1 — High priority

Own linked account exemption

- Valid linked account bypasses limits; unlinked accounts do not.
- Ownership validation and spoofing attempts.

Risk: *a gap here is a direct, exploitable limit-bypass vulnerability.*

Tier upgrade after KYC

- Upgrade effect on active sessions, new sessions, and in-flight transfers.
- Daily usage carry-over after upgrade.

Risk: *incorrect behaviour creates inconsistent customer experience and support escalations.*

9.3 P2 — Medium priority

Scheduled transfers

- Execution on schedule; insufficient remaining limit at execution; tier changes before execution; retry behaviour; failure notification.

Error messages

- Correct rejection reason, helpful guidance, appropriate error codes, localization.

9.4 P3–P4 — Low priority

Cosmetic, layout, and enhancement items are validated last, once higher-risk financial logic is confirmed stable.

10. Automation vs. Manual Testing

10.1 Automated testing

Automation is prioritized for repetitive, deterministic, high-risk scenarios executed frequently across regression cycles.

Feature	Why automate
Per-transaction limit validation	Large number of boundary combinations, run every build.
Daily limit calculations	Many data permutations; needs deterministic, repeatable checks.
Tier validation	Regression coverage across all tiers on every release.
API-level validation	Fast, reliable, and independent of UI changes.
Concurrent transfer scenarios	Parallel execution can be simulated consistently and repeatably.
Daily reset logic	Time-based regression tests (clock manipulation / time travel in test env).
Refund / reversal calculations	Accounting consistency must be verified identically across releases.
Scheduled transfer execution	Repeatable, time-based scenario, well suited to scheduled test runs.

10.2 Manual testing

Manual testing is reserved for exploratory, usability, and human-judgement scenarios that automation cannot reliably evaluate.

Feature	Why manual
User experience during KYC upgrade	Requires validating the real, end-to-end user flow.
Error message readability	Requires human judgement of clarity and tone.
Visual / UI verification	Layout, alignment, and responsiveness.
Exploratory testing	Discovering unexpected behaviours outside scripted paths.
Accessibility checks	Human evaluation of usability.
Edge cases found during testing	Investigating scenarios automation may miss.

11. Tier × Transfer Type × Limit-State Coverage Approach

Rather than writing 300–500 near-identical test cases, coverage is built from a small set of independent dimensions, combined using equivalence partitioning, boundary value analysis, and pairwise testing — maximizing business-rule coverage with a maintainable test count.

11.1 Test dimensions

Dimension	Values
Account Tier	Tier 1, Tier 2, Tier 3
Transfer Type	Wallet → Wallet, Wallet → Bank, Own Linked Account
Limit State	Below Limit, At Limit, Above Limit
Daily Usage	No Usage, Partial Usage, Daily Limit Reached
Transfer Status	Successful, Failed, Reversed, Pending
Timing	Normal, Just Before Reset, Just After Reset
Execution Type	Immediate, Scheduled

11.2 Prioritize by business risk

Not every combination carries equal risk. Core financial controls receive exhaustive combination coverage; lower-risk interactions receive pairwise coverage instead.

Exhaustive coverage (P0 dimensions)

- Tier × Transfer Type × Limit State
- Tier × Daily Usage
- Tier × Concurrent Transfers
- Tier × Daily Reset

11.3 Boundary value analysis

Rather than testing every possible amount, only the boundaries are tested — the same three-point pattern is reused for every tier.

Test amount (Tier 1, ₦20,000 limit)	Expected result
₦19,999	Allow
₦20,000	Allow
₦20,001	Reject

11.4 Pairwise testing for lower-risk combinations

Dimensions such as Tier, Transfer Type, Schedule, and Refund do not need exhaustive combination — pairwise testing ensures every pair of input values appears together at least once, giving broad interaction coverage without full permutation.

Tier	Transfer Type	Schedule	Refund
Tier 1	Wallet	Immediate	No
Tier 1	Bank	Scheduled	Yes
Tier 2	Wallet	Scheduled	No

Tier	Transfer Type	Schedule	Refund
Tier 2	Bank	Immediate	Yes
Tier 3	Wallet	Immediate	Yes
Tier 3	Bank	Scheduled	No

11.5 Data-driven parameterization

A single reusable automated test is driven by tier data, rather than one bespoke test per tier — reducing maintenance while increasing consistency.

Tier	Transaction limit	Daily limit	Expected
Tier 1	₦20,000	₦50,000	Pass
Tier 2	₦200,000	₦500,000	Pass
Tier 3	Unlimited	Unlimited	Pass

11.6 Exploratory testing focus

Automation verifies deterministic business rules; exploratory testing targets scenarios that are difficult to predict or script:

- Multiple devices sending transfers simultaneously
- Tier upgrade occurring during an active session
- Network interruption mid-transfer
- Scheduled transfer execution around midnight
- Rapid retries after a failed transaction
- Unexpected combinations of reversals, retries, and pending transfers

11.7 Example coverage matrix

Tier	Transfer Type	Below Limit	At Limit	Above Limit	Daily Limit	Reset	Concurrent
Tier 1	Wallet→Wallet	✓	✓	✓	✓	✓	✓
Tier 1	Wallet→Bank	✓	✓	✓	✓	✓	✓
Tier 1	Own Linked Acct	✓	✓	✓*	N/A*	✓	✓
Tier 2	Wallet→Wallet	✓	✓	✓	✓	✓	✓
Tier 2	Wallet→Bank	✓	✓	✓	✓	✓	✓
Tier 2	Own Linked Acct	✓	✓	✓*	N/A*	✓	✓
Tier 3	Wallet→Wallet	✓	N/A	N/A	✓	✓	✓
Tier 3	Wallet→Bank	✓	N/A	N/A	✓	✓	✓
Tier 3	Own Linked Acct	✓	N/A	N/A	N/A*	✓	✓

* Subject to clarification of business rules for own-linked-account exemptions (see Open Questions).

11.8 Coverage strategy summary

Technique	Purpose
Equivalence partitioning	Reduce redundant input combinations
Boundary value analysis	Validate critical limit thresholds
Coverage matrix	Ensure every business rule is exercised
Pairwise testing	Efficiently cover interaction combinations
Data-driven automation	Reuse one test across tiers and limits
Exploratory testing	Discover unexpected behaviours and edge cases

12. Entry and Exit Criteria

12.1 Entry criteria

- Open questions in Section 7 have documented answers from the product team, or explicit acceptance of a stated assumption.
- Test environment is available with configurable system clock / time-travel capability for reset and scheduling tests.
- Test data covering all three tiers, with associated KYC states, is provisioned.
- API/service endpoints for transfer creation, scheduling, and reversal are stable and reachable in the test environment.

12.2 Exit criteria

- 100% of P0 test cases executed and passed; zero open P0 defects.
- 100% of P1 test cases executed; open P1 defects have documented, accepted workarounds or a committed fix date.
- Coverage matrix (Section 11) fully executed for all tier × transfer-type × limit-state combinations marked exhaustive.
- Concurrency and race-condition test suite passes under load equivalent to expected peak traffic.
- All open questions in Section 7 are resolved and reflected in either the specification or an accepted risk log entry.

13. Roles and Responsibilities

Role	Responsibility
Test Manager	Owns this test plan, risk assessment, and go/no-go recommendation.
Test Analysts / Engineers	Design and execute test cases; author automated regression and concurrency suites.
Product Owner	Resolves open questions in Section 4; approves scope and acceptance criteria.
Development Team	Fixes defects by priority; supports test environment configuration.

Role	Responsibility
Compliance / Risk	Reviews P0 financial-control test results prior to release sign-off.

14. Defect Management

- Defects are triaged and assigned severity/priority aligned to the risk levels in Section 8.1 (a P0 feature defect is treated as a release blocker by default).
- Any defect that allows a limit to be exceeded, or a reversal to misallocate funds, is escalated immediately regardless of when it is found in the cycle.
- Defects arising from an unresolved open question (Section 7) are logged against the question, not just the symptom, so root-cause fixes address the ambiguity.

15. Risks to the Testing Effort

Risk	Impact	Mitigation
Open questions remain unanswered before test design starts	Test cases built on wrong assumptions; rework	Block detailed test-case authoring on Section 4 sign-off for P0 items
Test environment cannot manipulate server time reliably	Reset/scheduling tests cannot be executed deterministically	Require time-travel capability as an entry criterion
Concurrency defects are hard to reproduce	Race conditions ship to production undetected	Dedicated load-based concurrency suite, not just functional tests
Scope creep from downgrade/AML logic not yet defined	Effort diverted from confirmed P0/P1 scope	Track as out-of-scope until specification is extended

16. Deliverables

- This risk-based test plan (Word/PDF).
- Prioritized test case suite.
- Automated regression and concurrency test suites.
- Open-question log with product-team responses (Section 7), maintained until closure.